

1. **What are the three major security properties considered in a cellular system? Explain how cellular systems provide/improve the three security properties from 1G to 2G and then to 3G.**
 - a. The three major security properties are Authentication, Confidentiality, and Location Privacy. 1G, being an analog system, was un-encrypted, had a basic authentication system, and was fairly simple to identify the user's location using towers. 2G introduced better authentication methods like including SIM authentication keys, encrypted audio that required the authentication key, and a temporary ID to reinforce anonymity. 2G later upgraded their encryption algorithms to use COMP128-4. 3G further improved on 2G by adding integrity mechanisms, stronger algorithms for encryption, and enhanced authentication systems.
2. **What is a rogue base station in the cellular network? Describe how a rogue base station launches the man-in-the-middle attack. Describe the reason why 1G, 2G and 3G are or are not vulnerable to this attack, respectively.**
 - a. A rogue base station is a man in the middle attack where the attacker sets up a false base station that forces your device to communicate to the real base station through the false one, allowing the attacker to identify, track, or eavesdrop. All three generations are vulnerable to a rogue base station in some capacity. 1G is not encrypted at all, so a rogue base station would be able to listen to the audio without any issue. 2G is encrypted, however the encryption algorithm used is not complex enough to prevent it from being cracked. 3G, while more strongly encrypted, can still be cracked and leave your information vulnerable.
3. **Describe the message delivery of SMS in different scenarios, when the target user is 1) not reachable, 2) reachable but not active, and 3) active at BTS. Discuss the vulnerabilities of this message delivery mechanism.**
 - a. If the target user is unreachable, the HLR notifies the SMSC and the SMSC stores the message in a queue until the user is reachable. If the user is reachable but not active, the HLR notifies the SMSC which MSC to go to, which then asks the VLR where the user is. If the VLR responds no, then the message is sent to BTSs until the user's device responds with an acknowledgement. If the user is active, then the VLR responds with which BTS the user is at and the message is sent there. A vulnerability of this system is SMS DoS attacks, where an attacker can flood the network with more SMS messages than the network is designed to handle.
4. **How much SMS traffic is needed to flood a city, assuming the city has 50 cell towers, 3 sectors/tower and 12 SDCCH/sector. Each SDCCH supports 900 messages/hour, and each message is 2KB.**
 - a. 7.2 Gbps – Work in uploaded photos
5. **Understand the modulo-2 division used for CRC check and design a 4-bit CRC encoding/decoding system for a sender and a receiver with polynomial $x^4 + x^3 + x + 1$.**
 - a. The sender has a 6-bit message 101100 to send. Compute the CRC checksum and show the bit sequence that will be sent. Derive the CRC decoding results at the receiver and show that whether the CRC check is successful.
 - b. Repeat the CRC ending with an 8-bit message 11110011. Assume that the 5th bit is toggled during the transmission. Show whether this bit error can be detected at the receiver.

- a. a) 1011001011
 - b) the remainder was 0111, indicating that a bit was flipped before reaching the receiver.
Work in uploaded photos.
- 6. Consider a simplified version of WEP system. This WEP system calculates ICVs using a 3rd-degree CRC polynomial $x^3 + x + 1$ and generates the stream cipher key based on RC4. At time t_1 , the sender wants to send a message 1010011101 to the receiver and uses the WEP system to encrypt the message. Given that the stream cipher key K output by RC4 is 1001101100111 at time t_1**
 - 1) derive the CRC-encoded message M (i.e., msg + ICV);
 - 2) write down the WEP-encrypted data $E\{\text{msg} + \text{ICV}\}$;
 - 3) Show that the receiver can decode to get the correct message 1010011101.
 - a. 1) 1010011101100
 - 2) 0011110001011
 - 3) Proof in uploaded photos
- 7. a. Draw a block diagram that illustrates the encapsulation process.**
 - b. Describe the steps at the receiver end to recover the plaintext and perform the integrity check.
 - c. Draw a block diagram that illustrates part b.
 - a. a) diagram in uploaded photos
 - b) in order to recover the plaintext, the receiver first uses an XOR operation on the encrypted data using the shared key in order to restore the data. Then it uses the ICV to verify that the data has not been modified.
 - c) Diagram in uploaded photos
- 8. Does this mean that the ICV is protected from the bit flipping attack? Explain.**
 - a. The ICV is not protected against a bit flipping attack as it is still possible to alter the data, since the bit will get flipped back to whatever it was before. Additionally, it is still possible to identify bits that can be flipped without affecting whether the ICV checksum will be able to identify the message has been tampered with. This results in a loss or corruption of data.

4. 50 towers 3 sectors 12 gch 900 mps
 2 KB msg
 $50 \cdot 3 \cdot 12 \cdot 900 = 1,620,000$ messages/hr
 $450 \text{ m/s} \cdot 2 \text{ KB} = 900 \text{ KBps}$
 $900 \cdot 8 = 7,200 \text{ Kbps}$
7.2 Gbps

5. $x^4 + x^3 + 0x^2 + x + 1$
 1 0 1 1
 a. 1011001011
 b. remainder: 011
 11011 | 101100000000 | 11011 | 11110011011
 11011
 11010
 11011
 000010
 00000
 00100
 00000
 01000
 00000
 10000
 11011
 01101
 00000
 1010
 11011
 00011
 00000
 00111
 00009
 0111

6. $x^3 + 0x^2 + x + 1$
 1 0 1 1
 P1: 101001101000
 P2: 100110110011
 P3: 00000
 1011 | 101001101000 | 1011 | 101001101000
 1011
 0010
 0000
 0101
 0000
 1011
 1011
 0000
 0000
 0010
 0000
 0101
 0000
 1010
 1011
 0010
 0000
 0100
 0100
 0000
 1000
 1011
 011

